Business Recommendation Report - Retail Inventory Analytics

1. Portfolio Diversification to reduce Concentration Risk

Problem

The current portfolio of **123 vendors and 9.5K products** is highly concentrated. The **top 10 vendors contribute 65% of total revenue**, while the **top 20% of products (by sold quantity) drive 85% of sales, profit, and procurement**. This indicates strong dependence on a limited set of vendors and products, increasing supply and revenue risk.

At the same time, there is a large underutilized long tail, with several vendors and products not contributing proportionately to business performance.

Recommendation

- Expand procurement from **12 identified high-potential vendors** and optimize engagement with **49 existing vendors** to reduce dependency on top contributors
- Increase procurement focus on around **2.1K identified products** to strengthen depth in the portfolio and improve inventory distribution.
- Drive revenue growth by promoting **132 high-potential products** through targeted pricing, visibility, and demand generation initiatives.
- Maintain strategic partnerships and volume commitments with top vendors and products, while gradually reducing their overall share through controlled diversification.

2. Optimization of Inventory Investment Allocation

Problem

The current investment allocation across the portfolio is inefficient and misaligned with product performance. Around **85% of the procurement budget is concentrated on around 5.2K mid-performing products**, while **2.1K high performing products receive only 8%** of the budget. Additionally**, 7% of the budget is locked in 3.1K obsolete products**, leading to capital inefficiency.

From an inventory perspective, only **37% of capital is allocated to high-return products**, whereas a **significant 62% is tied up in moderate, low, or negative return products**, limiting overall profitability and capital productivity.

Recommendation

- Rebalance procurement budget by increasing allocation **to 2.1K high performing products** and reducing excess investment in mid performing products.

- Gradually phase out or liquidate **3.1K obsolete products to free up 7% of locked capital**.
- Optimize inventory allocation by shifting capital from low and negative return products towards high return categories.
- Implement performance based inventory planning, where procurement and stock levels are aligned with product profitability and turnover metrics.

3. Improving Inventory Movement and Turnover Efficiency

Problem

Inventory movement across the year has been inconsistent. Inventory turnover improved from 0.4 to 0.7 during growth months but declined again before recovering at year end, with DOI showing an inverse trend.

While inventory remained aligned with demand during growth phases, it adjusted slowly during demand slowdowns, leading to temporary overstocking. **Inactive inventory levels remained volatile (6.5%–9.2%),** indicating inconsistent stock utilization. Additionally, inventory value steadily increased with **higher variability in later months (CV rising from 0.4% to 3.4%)**, suggesting reduced control as stock levels grew.

From a product movement perspective, **12% fast moving products drive the majority of revenue and inventory**, while **53% slow moving products contribute only 3% revenue but still occupy 9% of inventory**, highlighting inefficiencies in stock allocation. Vendor performance is also uneven, with a portion of vendors contributing low returns and holding unsold inventory.

Recommendation

- Prioritize procurement and inventory allocation **towards 2.1K high efficiency and fast moving products** to improve turnover and revenue contribution.
- Strengthen engagement with **38 key vendors and optimize sourcing from 49 high-efficiency vendors** to ensure better inventory flow and responsiveness.
- Reduce exposure to **38 low performing vendors** by limiting procurement, clearing unsold inventory, and reallocating budget to high return segments.
- Align replenishment cycles dynamically with demand trends and integrate turnover, DOI, and sell-through metrics into ongoing inventory and vendor planning to minimize overstocking and variability.

4. Cost Optimization and Pricing Strategy Enhancement

Problem

The current pricing and cost structure indicates margin optimization opportunities across segments. While the budget segment with **around 4.2K products holds highest revenue share (46%) with strong margins (around 32%)**, the mid-range segment **has 3.7K products and contributes 38% revenue but operates at lower returns (around 30%)**. Premium and luxury segments have limited product count but underperform on margins, indicating pricing inefficiencies.

Although selling prices are consistently above purchase prices, significant overlap suggests tight margins across most products. Pack size analysis further shows imbalance as **standard pack size with 8.4K products has 56% revenue and healthy margin of 32%**, **while large packs has 1.2K products and contribute 36% revenue but with lower margins 28.4%**. **Small packs with around 700 products have decent margins (27.5%) but low scale (5.4% revenue**), and **bulk pack size with 163 products offer a strong margin of 38% but minimal contribution 2% revenue**, indicating underutilization. Additionally, profitability is not strongly linked to demand, highlighting pricing as a key lever.

On the cost side, bulk purchasing reduces unit cost but introduces higher inventory risk without proportional efficiency gains. A large portion of products and vendors also show clear opportunities for pricing correction and cost negotiation.

Recommendation

- Implement targeted pricing adjustments**: increase prices for around 3.2K identified products which holds 58% revenue share** and **reduce prices for 2,.3K low performing products having only 6% revenue share** to optimize demand and margins.
- Optimize pack size strategy by scaling high margin bulk and small packs selectively, improving margins in large packs, and sustaining strong performance of standard packs.
- Standardize markup strategy across pack sizes and segments, with tighter control on low margin categories, especially mid-range, premium, and luxury segments.
- Initiate **immediate cost negotiations with around 40 high impact vendors (52% revenue share) and optimize terms with 33 additional vendors**, while maintaining strong relationships with 33 key partners.
- Avoid excessive bulk procurement despite lower unit costs, instead balance order size with inventory efficiency to prevent overstocking and holding risk.
- Leverage pricing as a strategic driver by continuously aligning prices with product performance, ensuring high return products are scaled while underperforming ones are corrected or repositioned.